
Multimodal Models

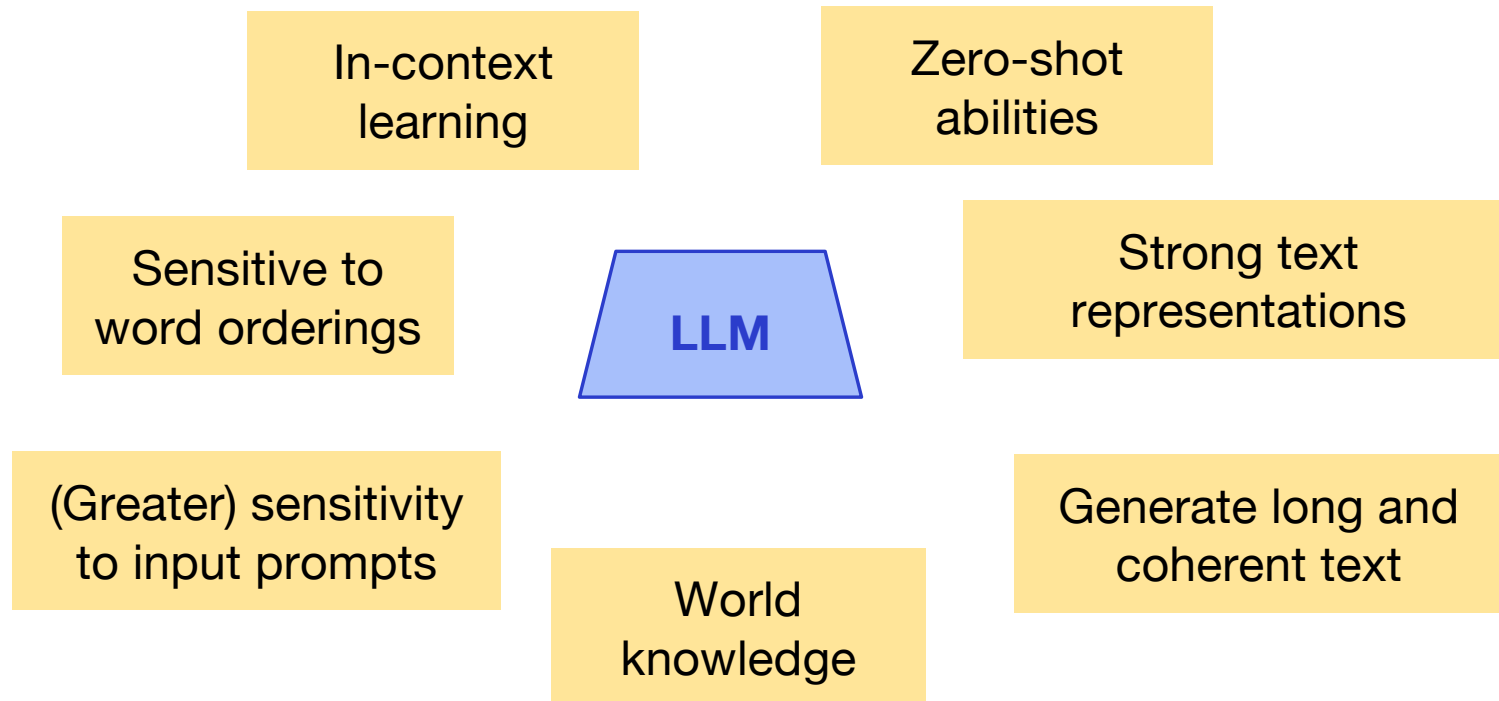
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Carnegie Mellon University

**Carnegie
Mellon
University**



Large Language Models



What is Missing?

- Image + text inputs → text outputs
 - Much more general than text-only LMs
- **Early on (2014 - 2020):** Finetune on paired image-text datasets for specific tasks
- **More recently:** Map a strong vision model to an LLM
 - Use prompting rather than finetuning
 - Leverage the abilities of pretrained LLMs

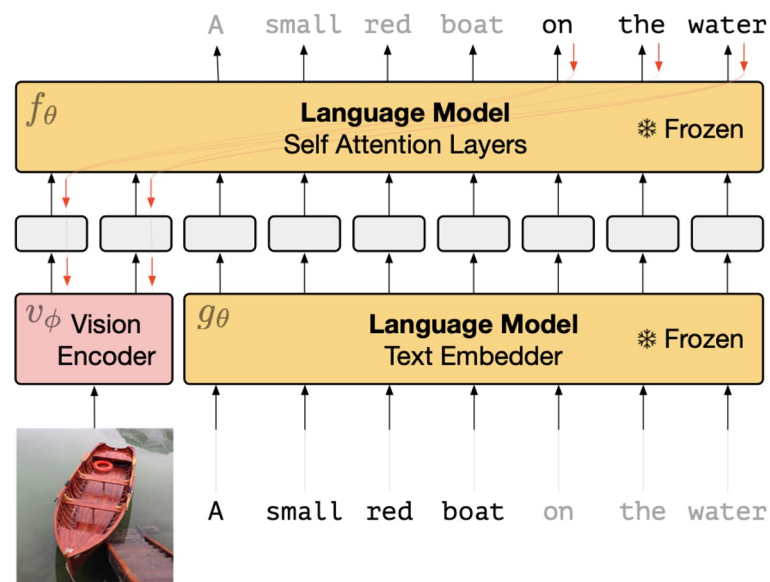


Image to Text Tasks

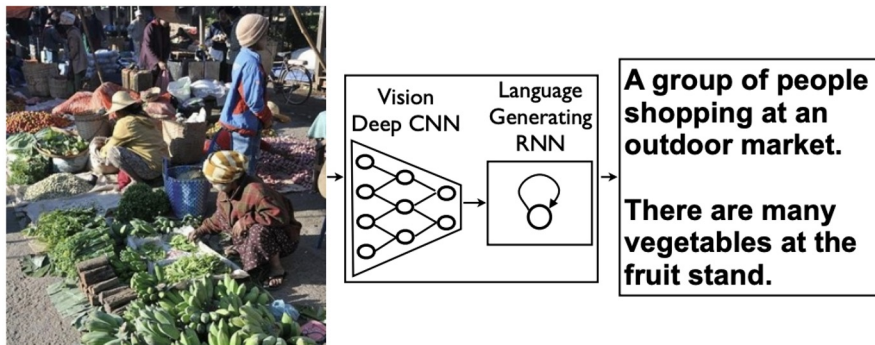


Image Captioning



What color are her eyes?
What is the mustache made of?



How many slices of pizza are there?
Is this a vegetarian pizza?



Is this person expecting company?
What is just under the tree?



Does it appear to be rainy?
Does this person have 20/20 vision?

VQA: Visual Question Answering

Show and Tell: A Neural Image Caption Generator ([Vinyals et al., 2014](#))

VQA: Visual Question Answering ([Agrawal et al., 2016](#))

Image to Text Tasks



Visual Dialog

Q: What is the gender of the one in the white shirt ?

A: She is a woman

Q: What is she doing ?

A: Playing a Wii game

Q: Is that a man to her right

A: No, it's a woman

Visual Dialogue

Visual Dialog ([Das et al., 2017](#))

Towards VQA Models That Can Read ([Singh et al., 2019](#))



What does it say near the star on the tail of the plane?

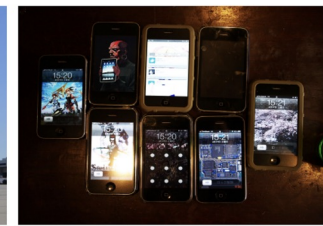
Ground Truth

jet

Prediction

nothing

(a)



What is the time on bottom middle phone?

Ground Truth

15:20

Prediction

12:00

(b)



What is the top oz?

Ground Truth

16

Prediction

red

(c)



What is the largest denomination on table?

Ground Truth

500

Prediction

unknown

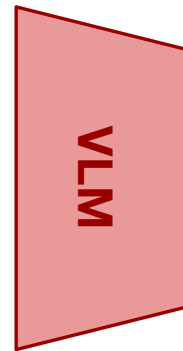
(d)

TextVQA

Image to Text Tasks Image Identification



Q: Please can you tell me what this item is?



A: butternut squash red pepper soup

Image-to-Text Applications: Sketch-to-Website

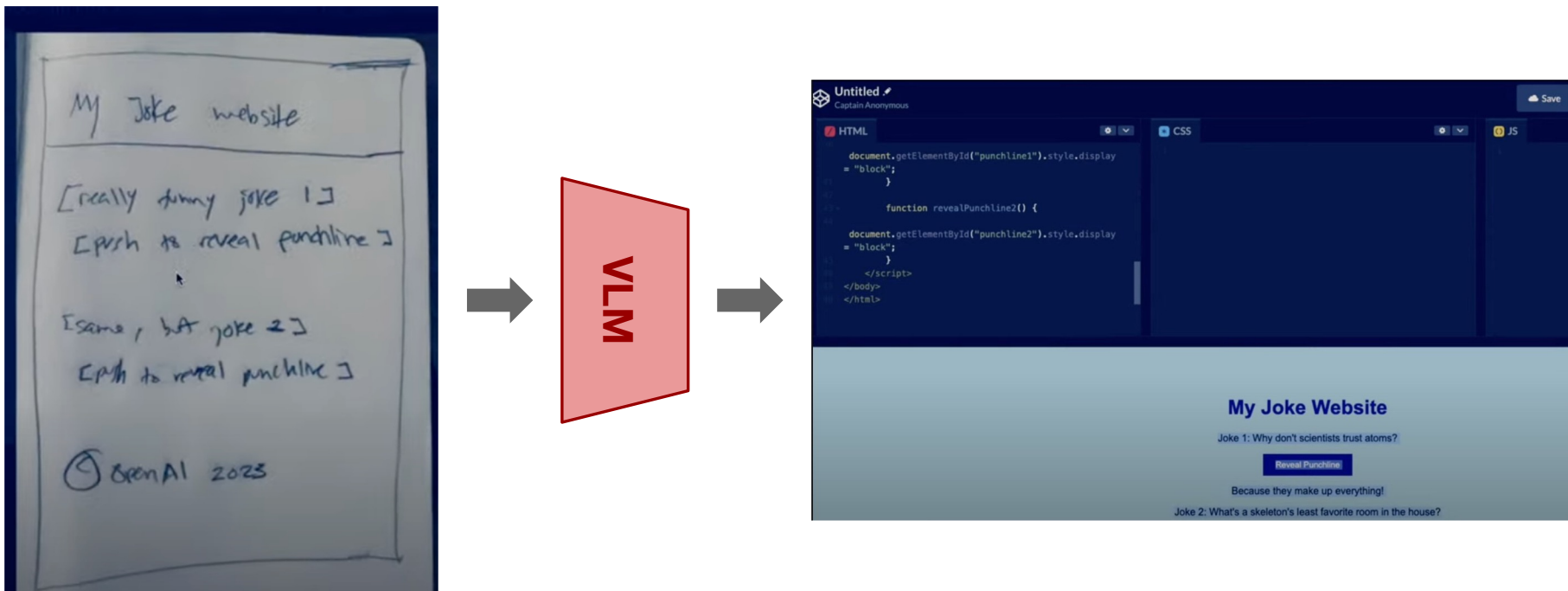


Image-to-Text Applications: Image to Latex

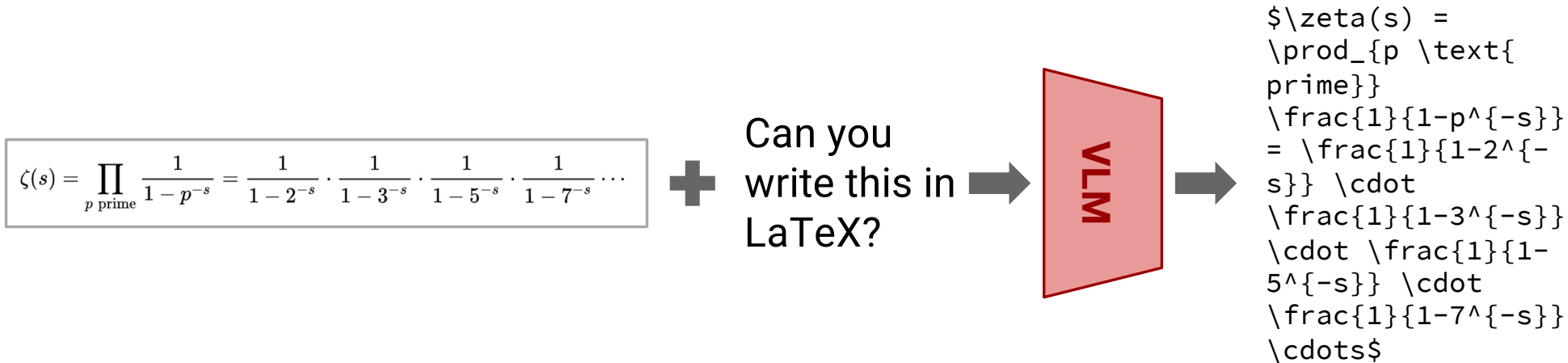


Image-to-Text Applications: Image to Latex

LaTeX formatting for Riemann zeta function Euler product formula ∨

$$\zeta(s) = \prod_{p \text{ prime}} \frac{1}{1 - p^{-s}} = \frac{1}{1 - 2^{-s}} \cdot \frac{1}{1 - 3^{-s}} \cdot \frac{1}{1 - 5^{-s}} \cdot \frac{1}{1 - 7^{-s}} \cdots$$

JY Can you write this in LaTeX?

Here is the LaTeX code for the given mathematical expression:

```
\zeta(s) = \prod_{p \text{ prime}} \frac{1}{1 - p^{-s}} = \frac{1}{1 - 2^{-s}} \cdot \frac{1}{1 - 3^{-s}}
\cdot \frac{1}{1 - 5^{-s}} \cdot \frac{1}{1 - 7^{-s}} \cdots
```

This expression represents the Euler product formula for the Riemann zeta function, where the product is taken over all prime numbers p . Each factor in the product is of the form $(1 - p^{-s})^{-1}$, where s is a complex variable.

Copy Retry Share

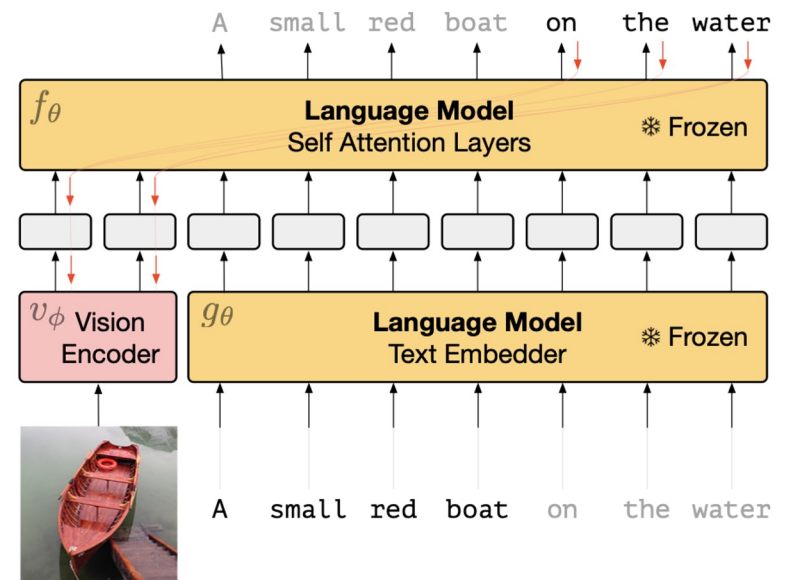


Claude can make mistakes. Please double-check responses.

How do you train VLMs?

Training Multimodal LLMs

- Image + text inputs → text outputs
 - Very general!
- **Early on (2014 - 2020):** Finetune on paired image-text datasets for specific tasks
- **More recently:** Staple a pretrained vision model to a pretrained LLM



LLMs are impressive general models

Chain of Thought Prompting

Input

Q: Roger has 5 tennis balls. He buys 2 more cans of tennis balls. Each can has 3 tennis balls. How many tennis balls does he have now?

A: Roger started with 5 balls. 2 cans of 3 tennis balls each is 6 tennis balls. $5 + 6 = 11$. The answer is 11.

Q: The cafeteria had 23 apples. If they used 20 to make lunch and bought 6 more, how many apples do they have?

Model Output

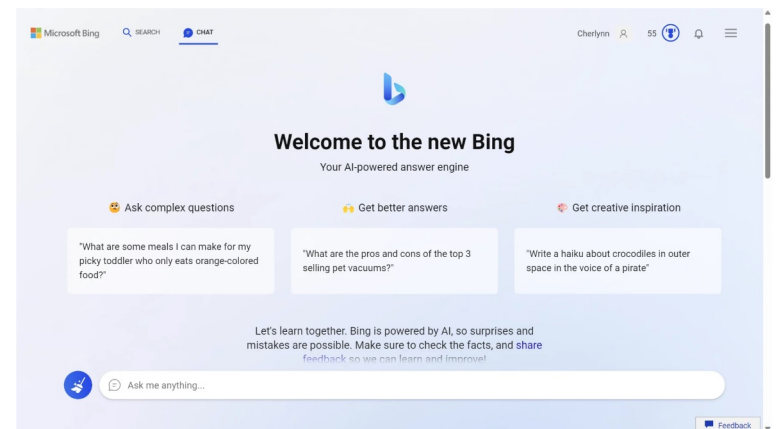
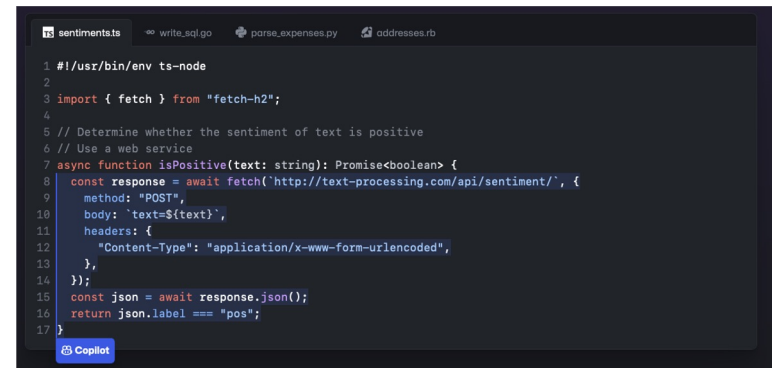
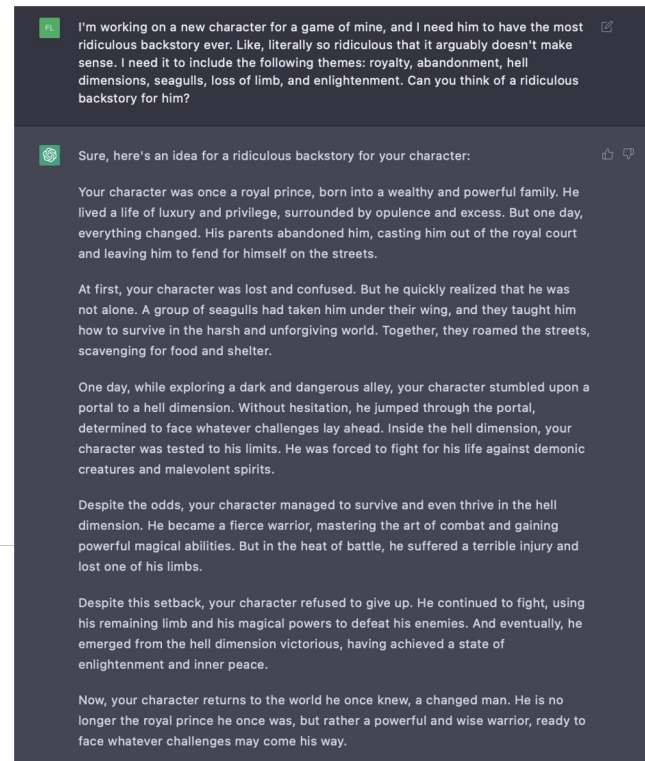
A: The cafeteria had 23 apples originally. They used 20 to make lunch. So they had $23 - 20 = 3$. They bought 6 more apples, so they have $3 + 6 = 9$. The answer is 9. ✓

(d) Zero-shot-CoT (Ours)

Q: A juggler can juggle 16 balls. Half of the balls are golf balls, and half of the golf balls are blue. How many blue golf balls are there?

A: **Let's think step by step.**

(Output) There are 16 balls in total. Half of the balls are golf balls. That means that there are 8 golf balls. Half of the golf balls are blue. That means that there are 4 blue golf balls. ✓



Chain of Thought Prompting Elicits Reasoning in Large Language Models ([Wei et al., 2022](#))

Large Language Models are Zero-Shot Reasoners ([Kojima et al., 2022](#))

ChatGPT ([OpenAI, 2022](#))

Copilot ([GitHub, 2021](#))

Bing ([Microsoft, 2023](#))

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I'm working on a new character for a game of mine, and I need him to have the most ridiculous backstory ever. Like, literally so ridiculous that it arguably doesn't make sense. I need it to include the following themes: royalty, abandonment, hell dimensions, seagulls, loss of limb, and enlightenment. Can you think of a ridiculous backstory for him?

Sure, here's an idea for a ridiculous backstory for your character:

Your character was once a royal prince, born into a wealthy and powerful family. He lived a life of luxury and privilege, surrounded by opulence and excess. But one day, his parents abandoned him, casting him out of the royal court and into the streets. He was forced to fend for himself, and he quickly realized that he was in a world of darkness and danger. He was scavenging for food and shelter, and he was constantly being hunted by creatures and malevolent spirits.

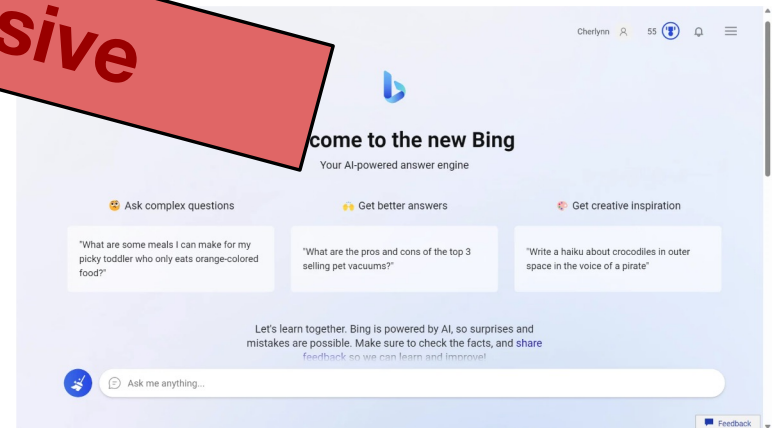
One day, while exploring a dark and mysterious portal to a hell dimension, he determined to face whatever challenges lay ahead. Inside the hell dimension, he was tested to his limits. He was forced to fight for his life against powerful magical abilities. But in the heat of battle, he suffered a terrible injury and lost one of his limbs.

Despite the odds, your character managed to survive and even thrive in the hell dimension. He became a fierce warrior, mastering the art of combat and gaining powerful magical abilities. But in the end, he emerged from the hell dimension victorious, having achieved a state of enlightenment and inner peace.

Now, your character returns to the world he once knew, a changed man. He is no longer the royal prince he once was, but rather a powerful and wise warrior, ready to face whatever challenges may come his way.

```
1 #!/usr/bin/env ts-node
2
3 import { fetch } from "fetch-h2";
4
5 // Determine whether the sentiment of text is positive
6 // Use a web service
7 async function isPositive(text: string): Promise<boolean> {
8   const response = await fetch("http://text-processing.com/api/sentiment/", {
9     method: "POST",
10    body: `text=${text}`,
11    headers: {
12      "Content-Type": "application/x-www-form-urlencoded",
13    },
14  });
15  const json = await response.json();
16  return json.label === "pos";
17 }
```

Resource Intensive



Chain of Thought Prompting Elicits Reasoning in Large Language Models ([Wei et al., 2022](#))

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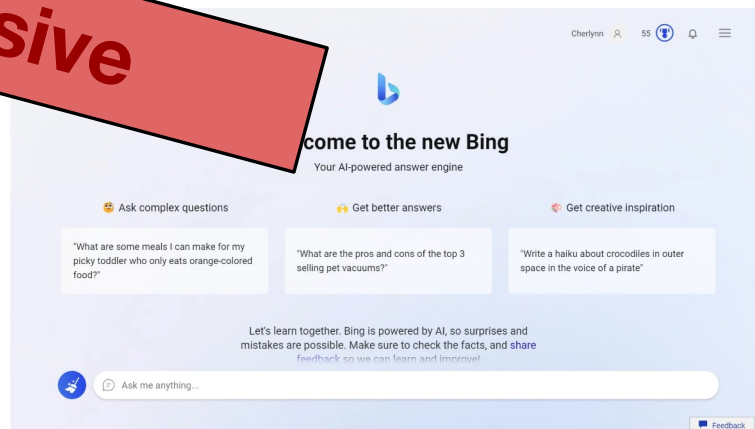
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9     method: "POST",
10    body: JSON.stringify({ text }),
11    headers: { "Content-Type": "application/json" },
12  });
13  const json = await response.json();
14  return json.sentiment === "positive";
15 }
```

Resolving Text → Text



Chain of Thought Prompting Elicits Reasoning in Large Language Models (Wei et al., 2022)

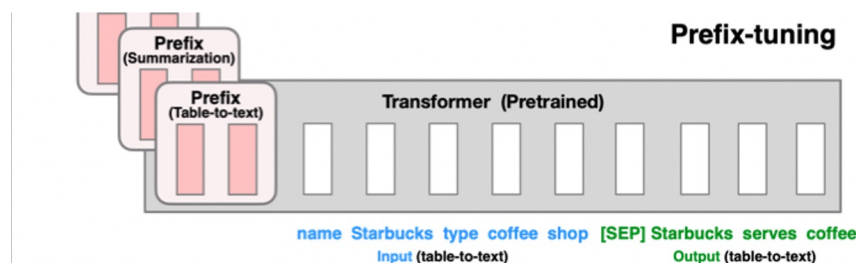
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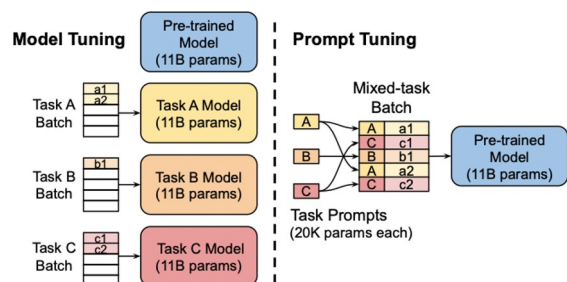
Copilot (GitHub, 2021)

Bing (Microsoft, 2023)

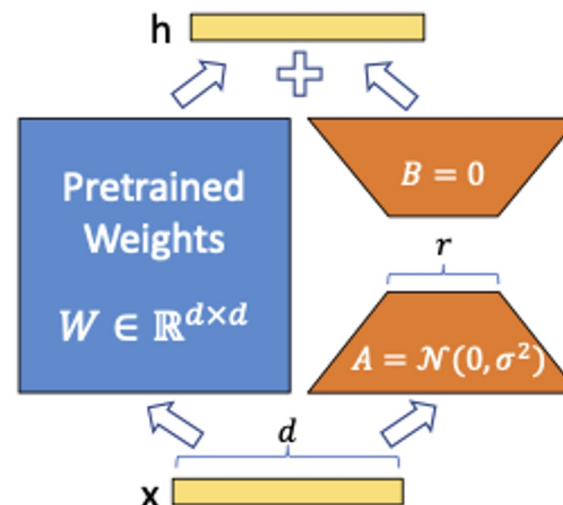
Parameter Efficient Adaptation



Prefix Tuning: Learns a prefix embedding (for each layer) to adapt to new tasks. ~99.9% of the model kept frozen.



Prompt Tuning: Similar idea to prefix-tuning, but learns just a single prefix for input embeddings.



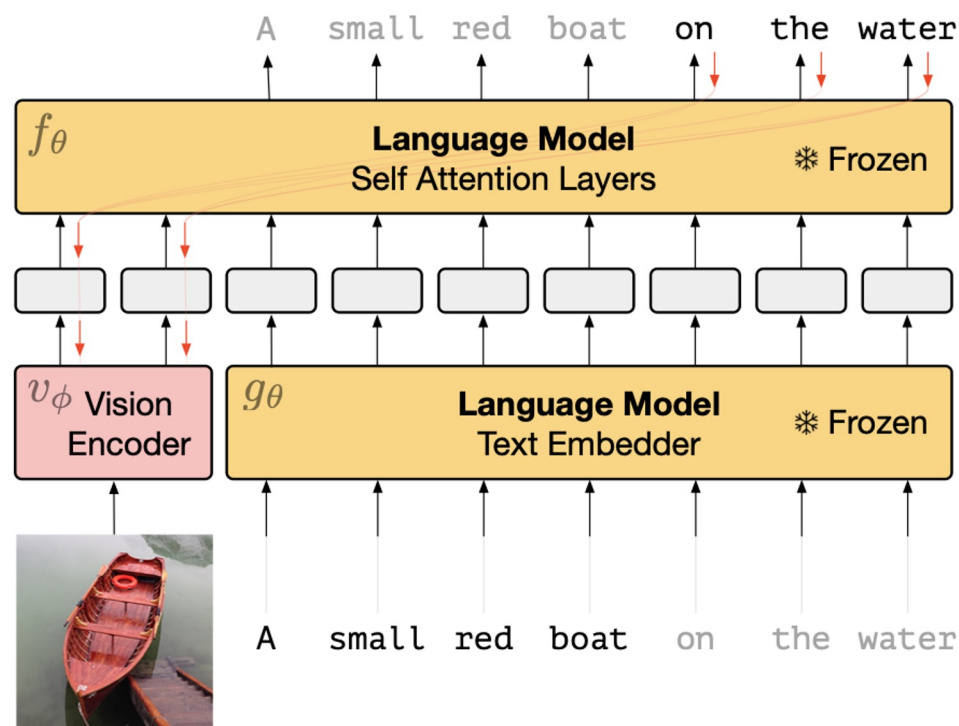
Low-Rank Adaptation (LoRA): Injects trainable rank decomposition matrices into each Transformer layer of a pretrained model.

Prefix-Tuning: Optimizing Continuous Prompts for Generation ([Li and Liang, 2021](#))
 The Power of Scale for Parameter-Efficient Prompt Tuning ([Lester et al., 2021](#))
 LoRA: Low-Rank Adaptation of Large Language Models ([Hu et al., 2021](#))

Multimodal LMs: Frozen (2021)

- Frozen**

- Prefix tuning for adapting LLMs to image captioning
- ~95% of the model kept frozen.
- Capable of compelling few-shot multi-modal reasoning



Multimodal LMs: Frozen (2021)

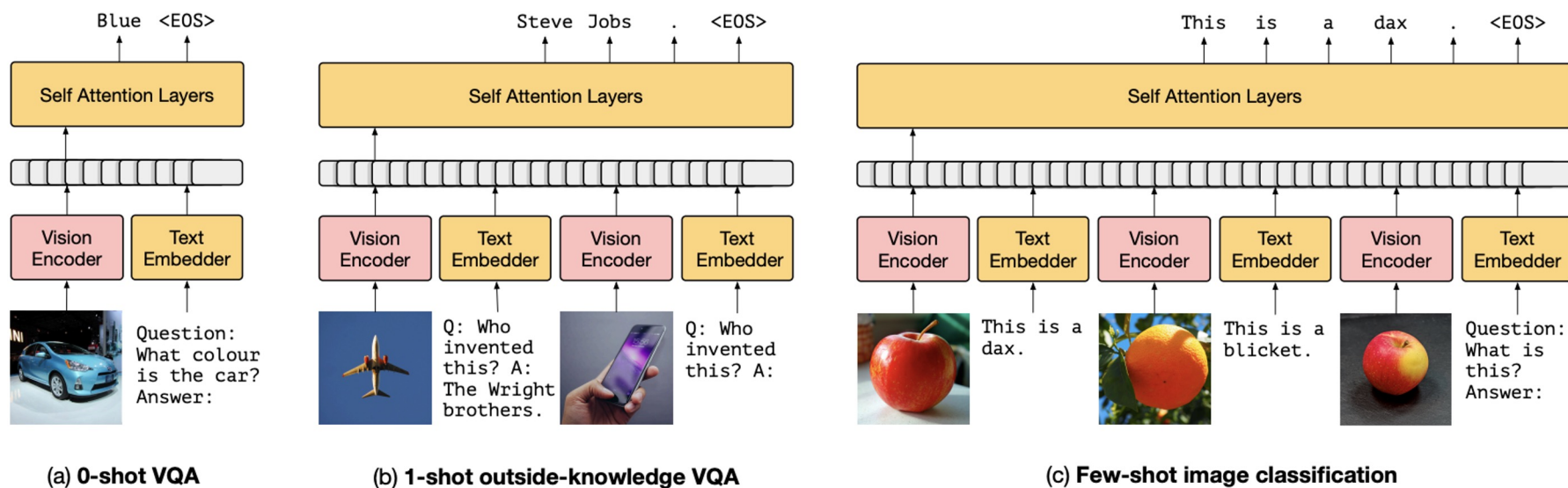


Figure 3: Inference-Time interface for *Frozen*. The figure demonstrates how we can support (a) visual question answering, (b) outside-knowledge question answering and (c) few-shot image classification via in-context learning.

 Frozen model  Mapping  Loss

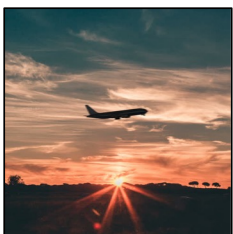
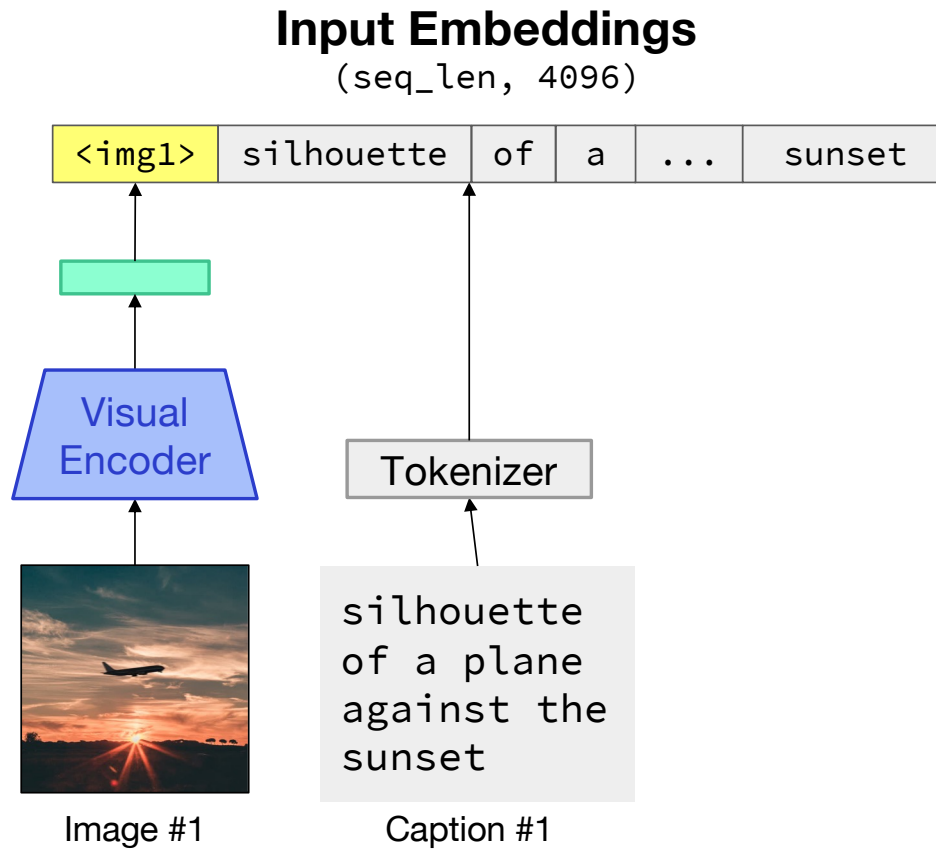


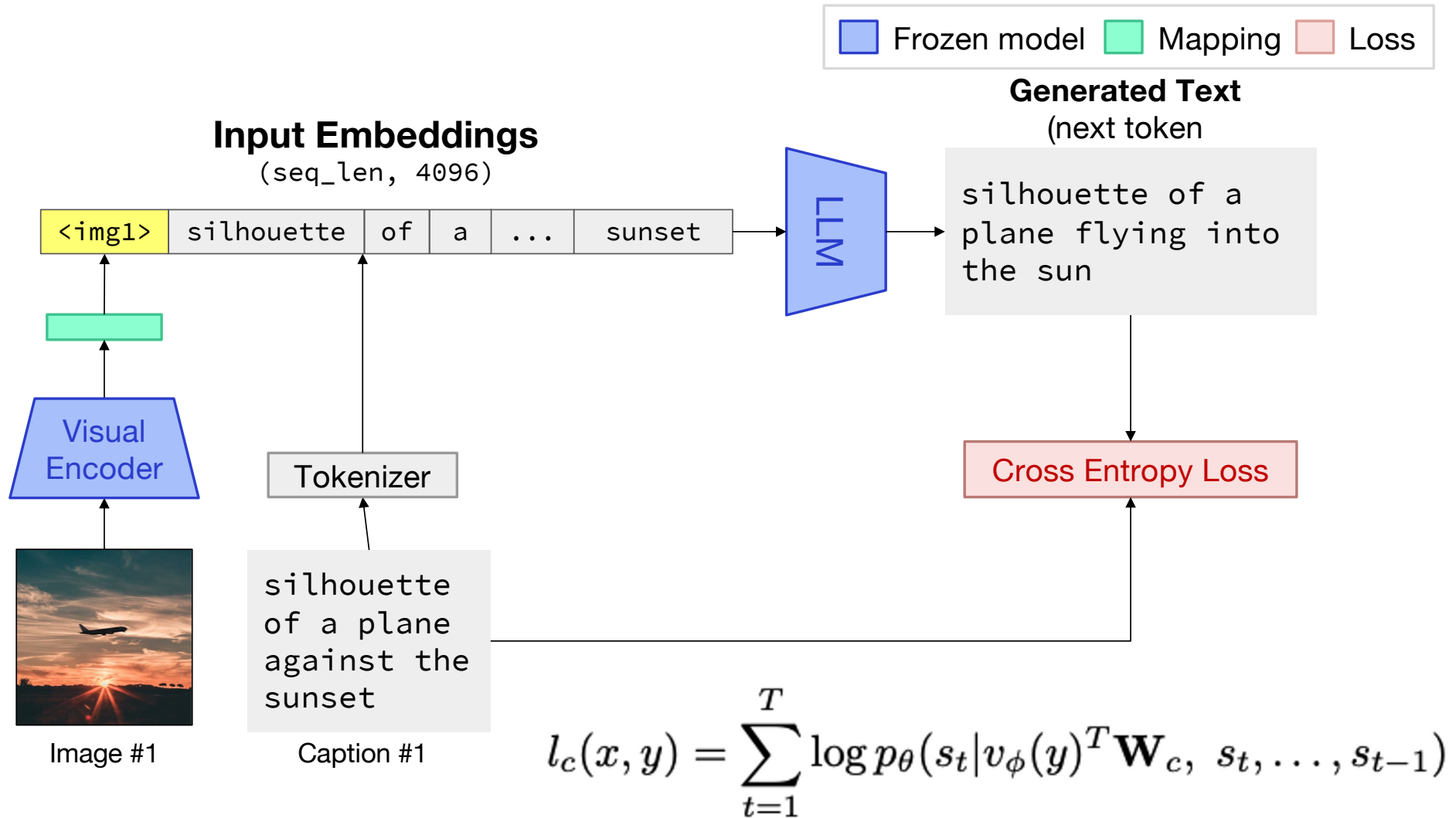
Image #1

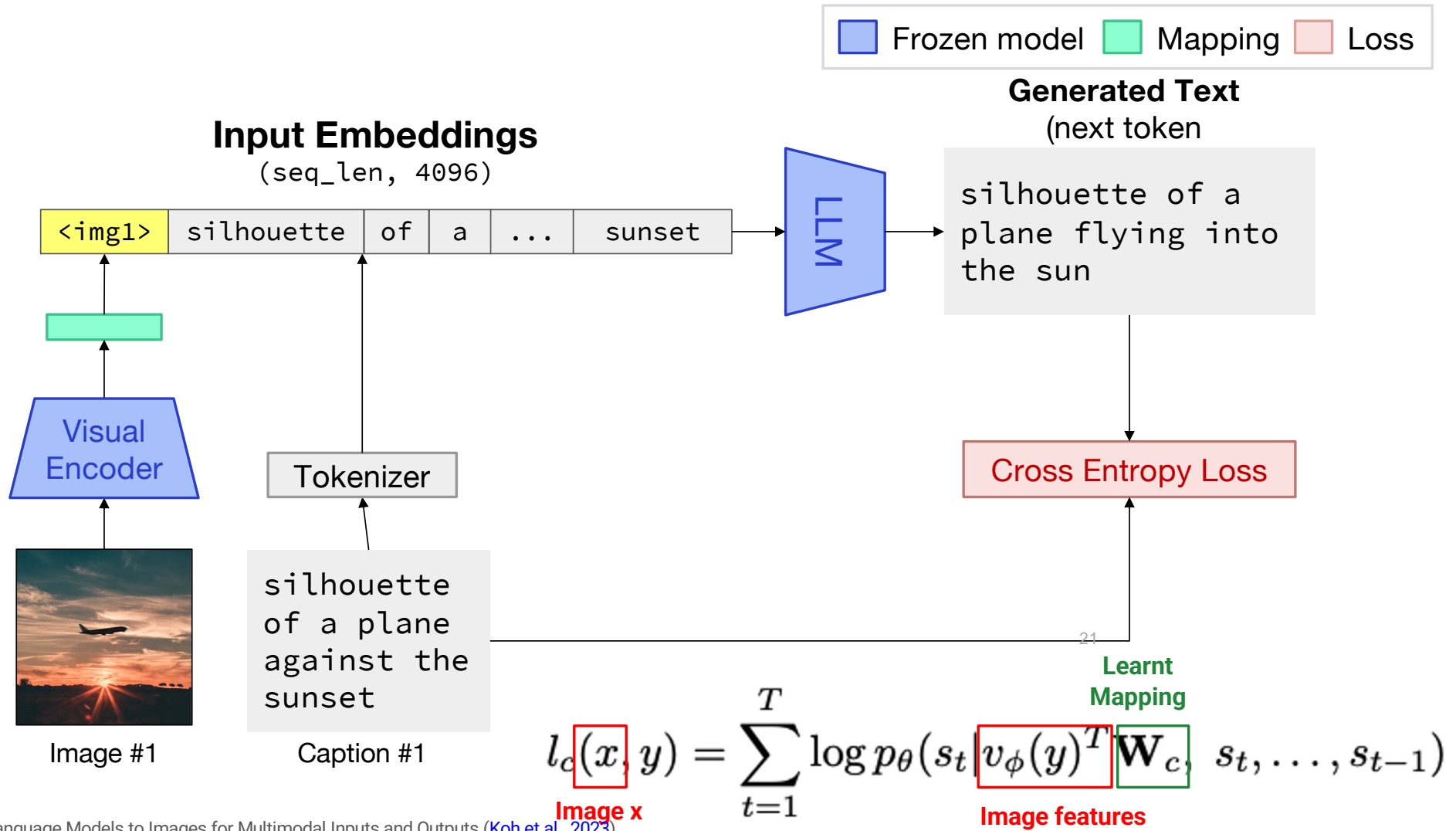
silhouette
of a plane
against the
sunset

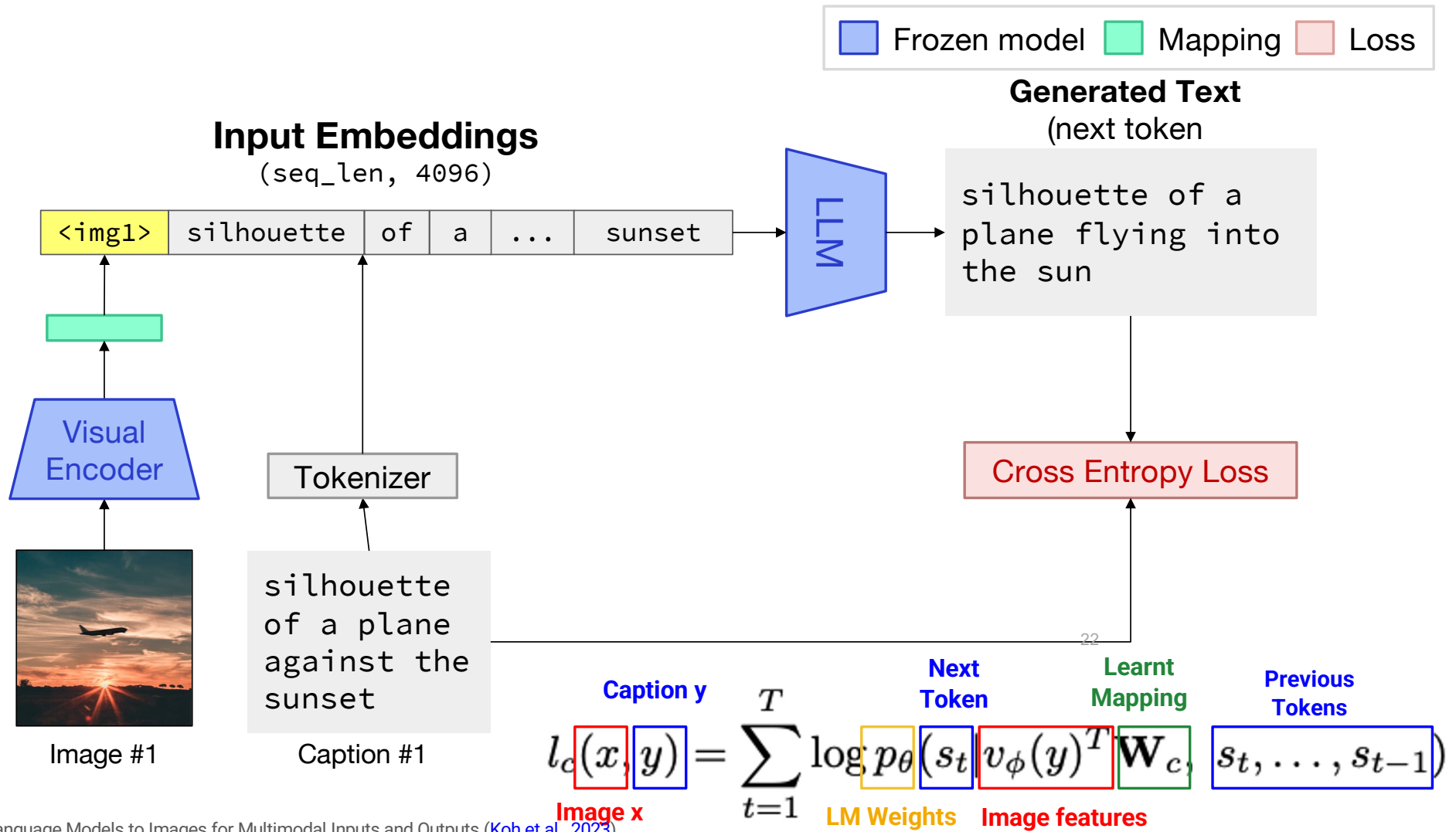
Caption #1

■ Frozen model ■ Mapping ■ Loss









How visually grounded are text-only LLMs?

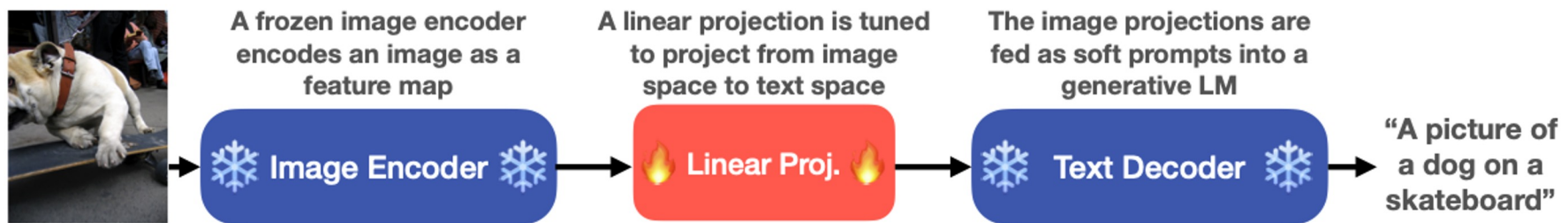
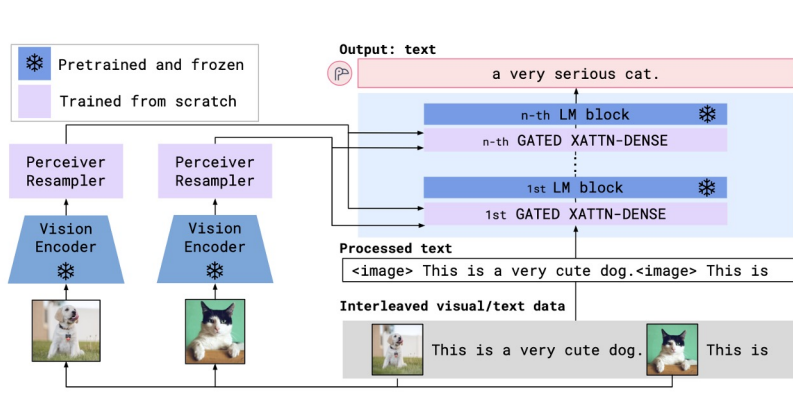


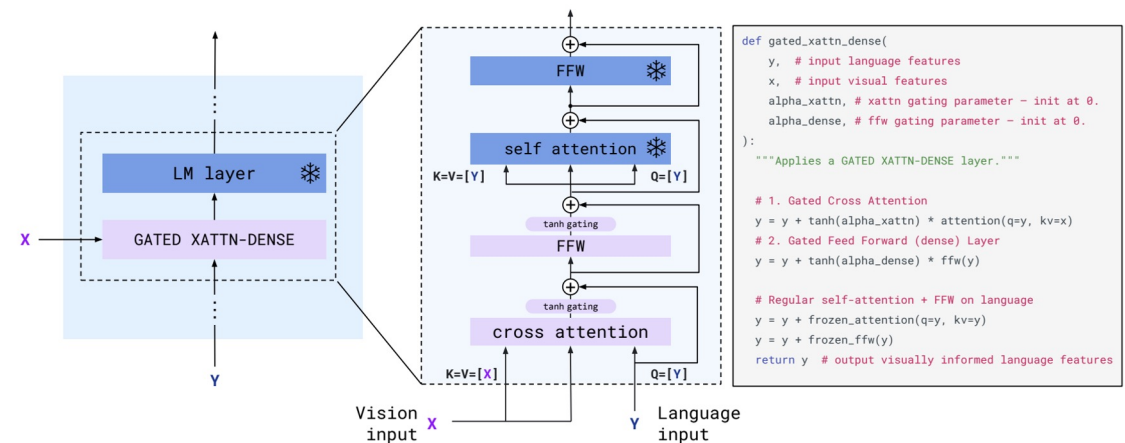
Figure 1: We train linear projections from image representations into the input space of a language model to produce captions describing images. We find that LMs can describe the contents of most image representations, but performance varies based on the type of image encoder used.

Merullo et al. showed that pretrained text-only LMs and pretrained visual encoders produce functionally equivalent representations up to a linear mapping.

Multimodal LMs: Flamingo

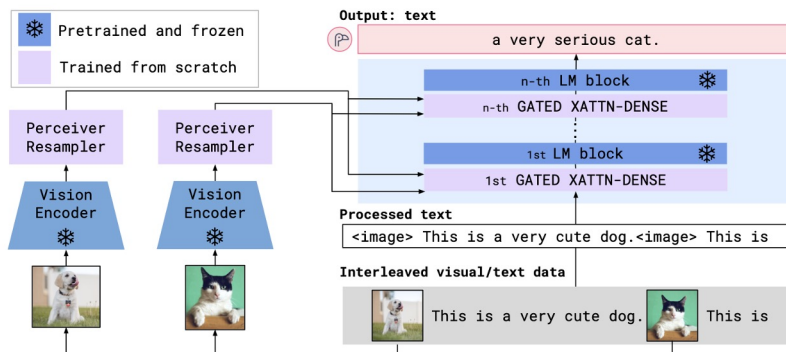


Flamingo: Finetunes new cross-attention layers on top of a 70B LLM. Achieves SOTA on many multi-modal tasks.

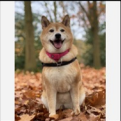
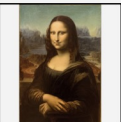
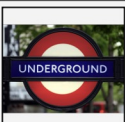
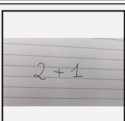
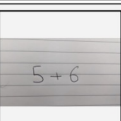
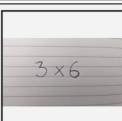
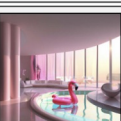
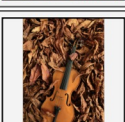
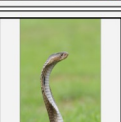
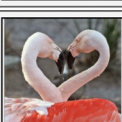


Introduced cross-attention layers between existing frozen LLM layers. Purple blocks are finetuned, blue blocks are kept frozen.

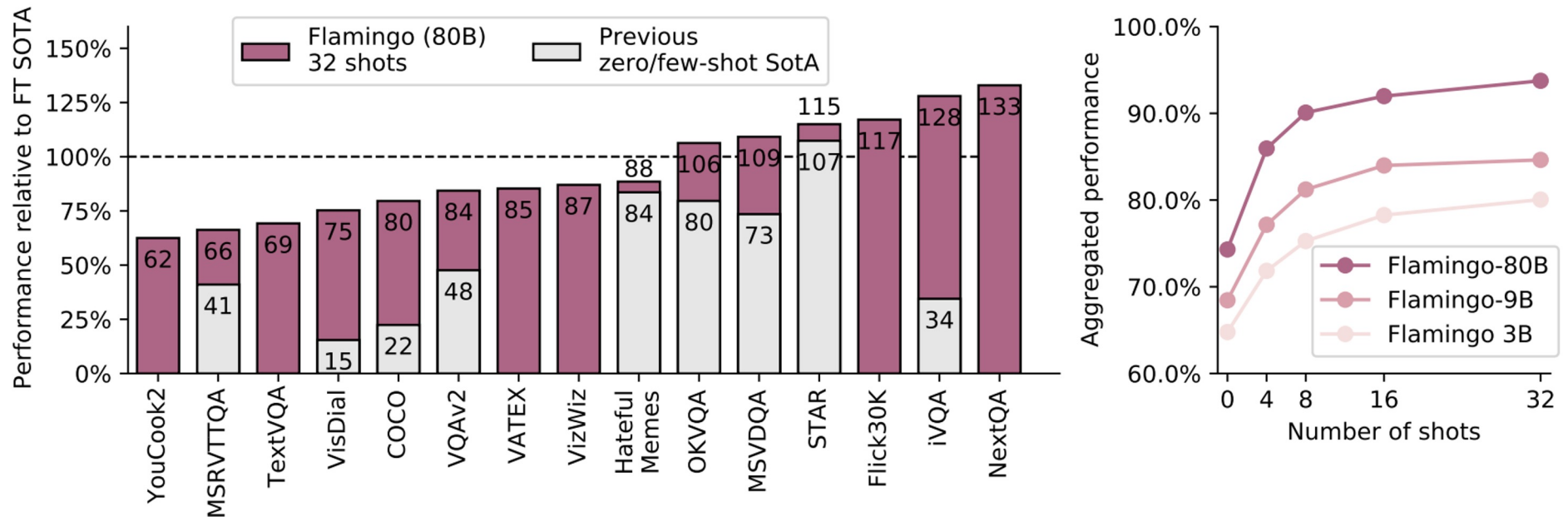
Multimodal LMs: Flamingo



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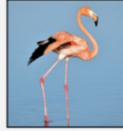
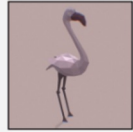
Input Prompt					Completion
	This is a chinchilla. They are mainly found in Chile.		This is a shiba. They are very popular in Japan.	 This is	a flamingo. They are found in the Caribbean and South America.
	What is the title of this painting? Answer: The Hallucinogenic Toreador.		Where is this painting displayed? Answer: Louvres Museum, Paris.	 What is the name of the city where this was painted? Answer:	Arles.
	Output: "Underground"		Output: "Congress"	 Output:	"Soulomes"
	2+1=3		5+6=11		3x6=18
	Output: A propaganda poster depicting a cat dressed as French emperor Napoleon holding a piece of cheese		Output: A pink room with a flamingo pool float.	 Output:	A portrait of Salvador Dali with a robot head.
	Les sanglots longs des violons de l'automne blessent mon coeur d'une langueur monotone.		Pour qui sont ces serpents qui sifflent sur vos têtes?		Je suis un cœur qui bat pour vous.

Multimodal LMs: Flamingo



Flamingo: a Visual Language Model for Few-Shot Learning ([Alayrac et al., 2022](#))

Multimodal LMs: Flamingo

 This is a picture of two teddy bears on the moon. What are they doing? They are having a conversation. What object are they using? It looks like a computer. Is this surprising? Yes, it is surprising. Why is this picture surprising to you? I think it is surprising because teddy bears are not usually found on the moon.	   What is the common thing about these three images? They are all flamingos. What is the difference between these three images? The first one is a cartoon, the second one is a real flamingo, and the third one is a 3D model of a flamingo.	 This is an apple with a sticker on it. What does the sticker say? The sticker says "iPod". Where is the photo taken? It looks like it's taken in a backyard. Do you think it is printed or handwritten? It looks like it's handwritten. What color is the sticker? It's white.	 This is a cityscape. It looks like Chicago. What makes you think this is Chicago? I think it's Chicago because of the Shedd Aquarium in the background.  What about this one? Which city is this and what famous landmark helped you recognise the city? This is Tokyo. I think it's Tokyo because of the Tokyo Tower.
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Data: Flamingo



Image-Text Pairs:

1.8B pairs with short captions (~12 tokens each)

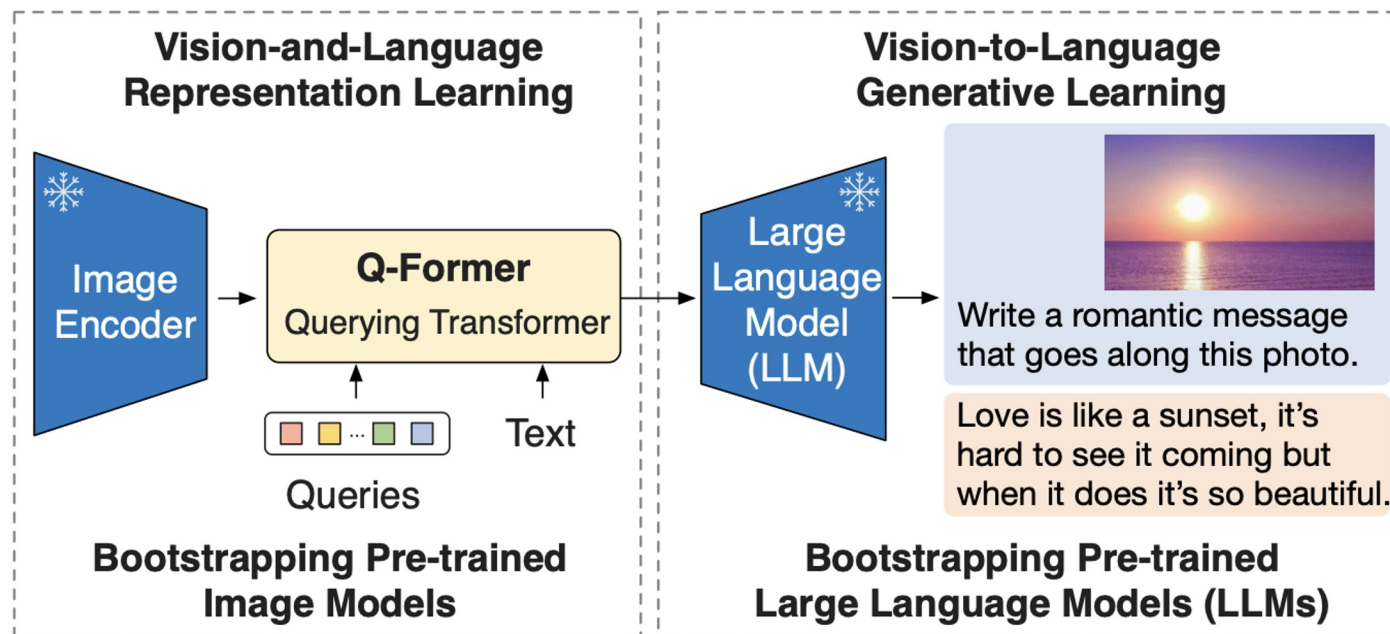
312M pairs with long descriptions (~20.5 tokens each)

Video-Text Pairs: 27M short videos

28

M3W: 43M webpages (for each page, sample 256 tokens, take first 5 images)

Multimodal LMs: BLIP-2

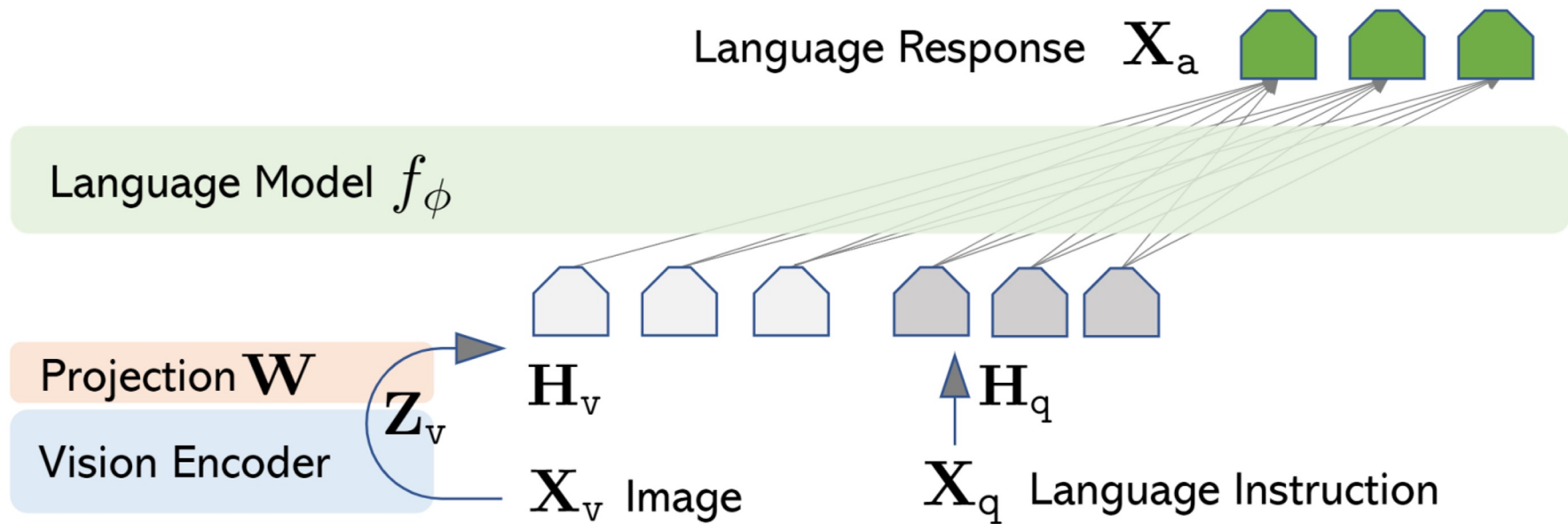


BLIP-2: Trains a Q-Former (essentially a transformer with learnt queries) to map between a frozen image encoder and a frozen LLM. Achieves strong performance on image captioning, VQA, and other image-to-text tasks.

Multimodal LMs: BLIP-2

 Explain the advantages of this product. The audi e-tron quattro concept is a plug-in hybrid electric sports car that has a range of 310 miles and can accelerate from 0 to 60 mph in just 4 seconds.	 Tell me something about the history of this place. The great wall of china was built in 221 bc by emperor qin shihuang to protect the emperor's capital from invaders from the north.	 Write down the facts that you know about this flower. It is a genus of flowering plants in the orchid family, orchidaceae. It is native to eastern asia and south east asia. It is a perennial flowering plant that grows up to a height of 1 m.
 Is this photo unusual? Yes, it's a house that looks like it's upside down. How could someone get out of the house? It has a slide on the side of the house.	 What are shown in the photo? A man and a chicken. What does the man feel and why? He is scared of the chicken because it is flying at him.	 What are the ingredients I need to make this? Pepperoni, mozzarella cheese, pizza sauce, olive oil, salt, pepper, basil. What is the first step? Place the pizza dough on a baking sheet, brush with olive oil, sprinkle with salt, pepper, and basil.

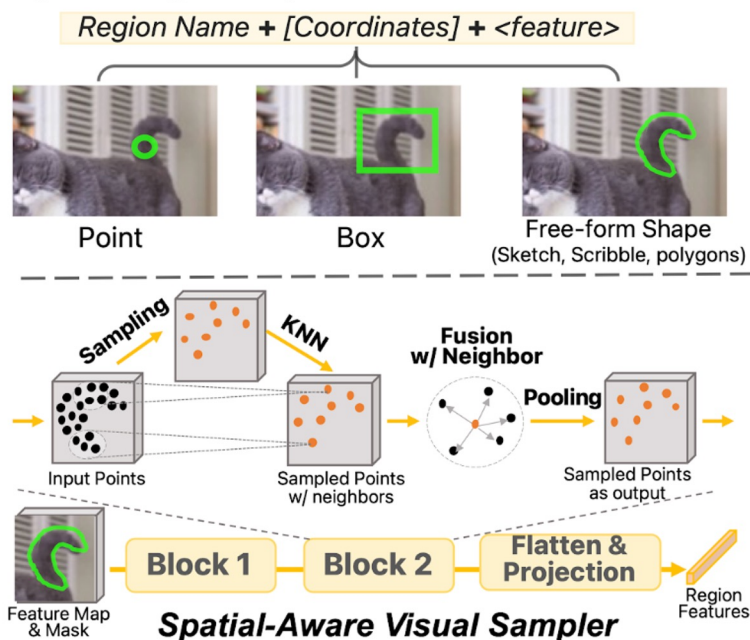
Multimodal LMs: LLaVA



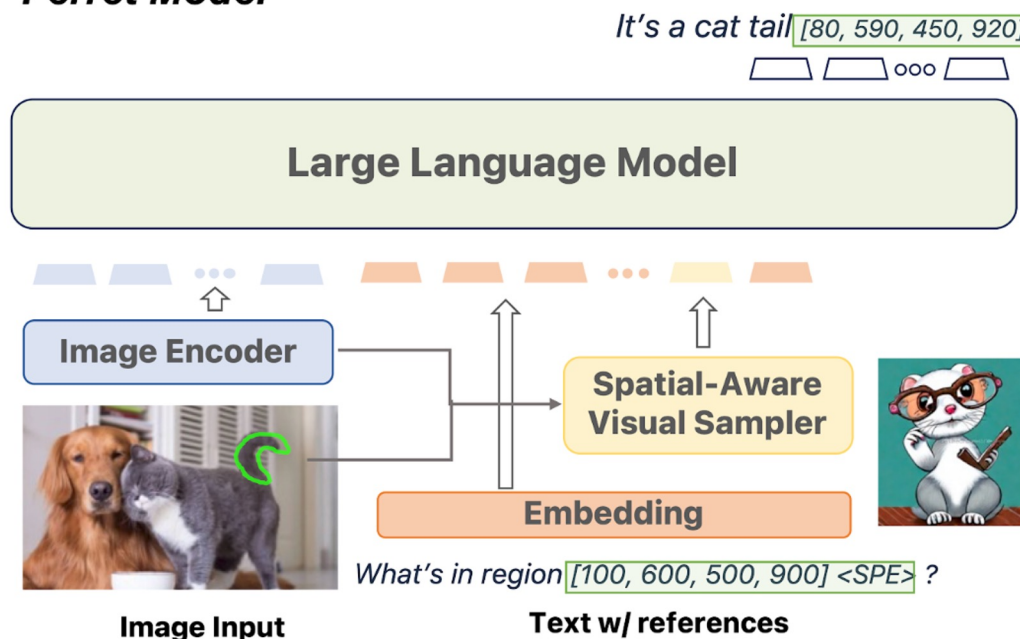
LLaVA: Finetunes a linear layer (W) over a frozen vision encoder and a frozen LLM. Showcases strong performance by finetuning on paired data of images and text instructions (some GPT-4 generated).

Multimodal LMs: Ferret

Hybrid Region Representation

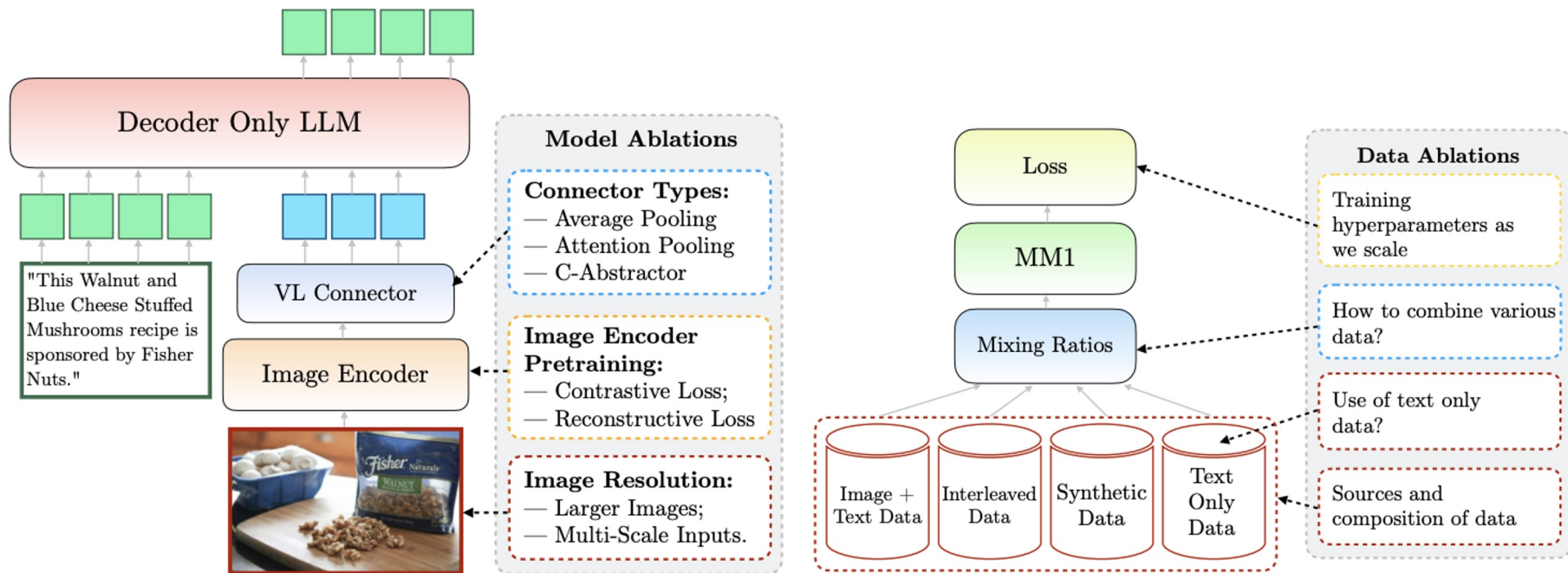


Ferret Model



Ferret: Hybrid input representations, enabling fine-grained grounding and referencing.

Multimodal LMs: MM1



MM1: ViT-L image encoder + C-Abstractor (CNN) mapper + 1.2B decoder-only LM

Multimodal LMs: MM1 Analysis

- **Biggest impact on results:**
 - Input image resolution has the highest impact (224px → 336px leads to +3% boost)
 - Model size (+1%) and training data composition (+1%) are also impactful
- **Vision-Language Connector:**
 - Number of visual tokens and image resolution matters most
 - The particular architecture has little effect
- **Pretraining Data:**
 - Interleaved data is instrumental for few-shot and text-only performance
 - Caption data lifts zero-shot performance
 - Text-only data helps with few-shot and text-only performance
 - Synthetic captions help with few-shot learning

Multimodal LMs: Others

- **Closed models:** GPT-5, Gemini, Claude 4
- **Open sourced models:**
 - Llama Family
 - Gemma
 - OpenFlamingo
 - Qwen-VL
 - CogVLM
 - and more each day...

Can we ground text-only LLMs to
consume and produce visual data?

GILL: Generating Images with Large Language Models



How can I make this more nutritious?

You can add vegetables to your ramen noodles, but you should be careful not to overdo it.



What are some vegetables I can add to it?

Broccoli, carrots, and green beans are all good choices.

I'm looking for some ideas for a new tattoo. What do you think looks good on a petite person?



I want a tattoo of a sunflower, but I'm not sure what style would look best on me.



That's beautiful. Show me one that would work well as a complementary piece





How should I publicize these at the market?

I would suggest a simple sign with the name of the business and a picture of the cupcakes.

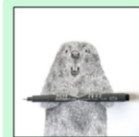
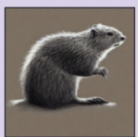


User prompts
 Retrieved
 Generated

GILL reference

GILL: A More General Multimodal LM

☐ User prompts ☐ Retrieved ☐ Generated

I am thinking of an animal. It is brown and furry, and has a tail. What is it? A squirrel. No, the animal I am thinking of lives in the river. A beaver. Show me an illustration of a beaver A beaver.  Yes, what about a pencil drawing of one?  What about a comic illustration of one? 	I am thinking of an animal. It is brown and furry, and has a tail. What is it? A squirrel. No, the animal I am thinking of lives in the river. A beaver. Show me an illustration of a beaver A beaver.  Yes, what about a pencil drawing of one?  What about a comic illustration of one? 	I am thinking of an animal. It is brown and furry, and has a tail. What is it? A squirrel. No, the animal I am thinking of lives in the river. A beaver. Show me an illustration of a beaver A beaver.  Yes, what about a pencil drawing of one?  What about a comic illustration of one? 
Ours <u>Retrieval and generation</u> multimodal LM. Decides when to retrieve or generate.	FROMAGE <u>Retrieval only</u> multimodal language model. Image outputs limited to the candidate retrieval set.	Stable Diffusion <u>Generation only</u> text-to-image model. Less sensitive to longer text inputs (such as dialogue).

GILL: A More General Multimodal LM

- **Frozen** (Tsimpoukelli et al., 2021)
Flamingo (Alayrac et al., 2022)
BLIP-2 (Li et al., 2023)
 - Process **image** + **text**, generate **text** only
- **FROMAGe** (Koh et al., 2023)
 - Process **image** + **text**, generate **text** + **retrieve images**
- **GILL**
 - Process **image** + **text**, generate **text** + **retrieve images** + **generate images**
 - Decides whether to retrieve images or generate from scratch
 - Resource efficient: trained on 2 GPUs for 2 days

Multimodal Few-Shot Learning with Frozen Language Models ([Tsimpoukelli et al., 2021](#))

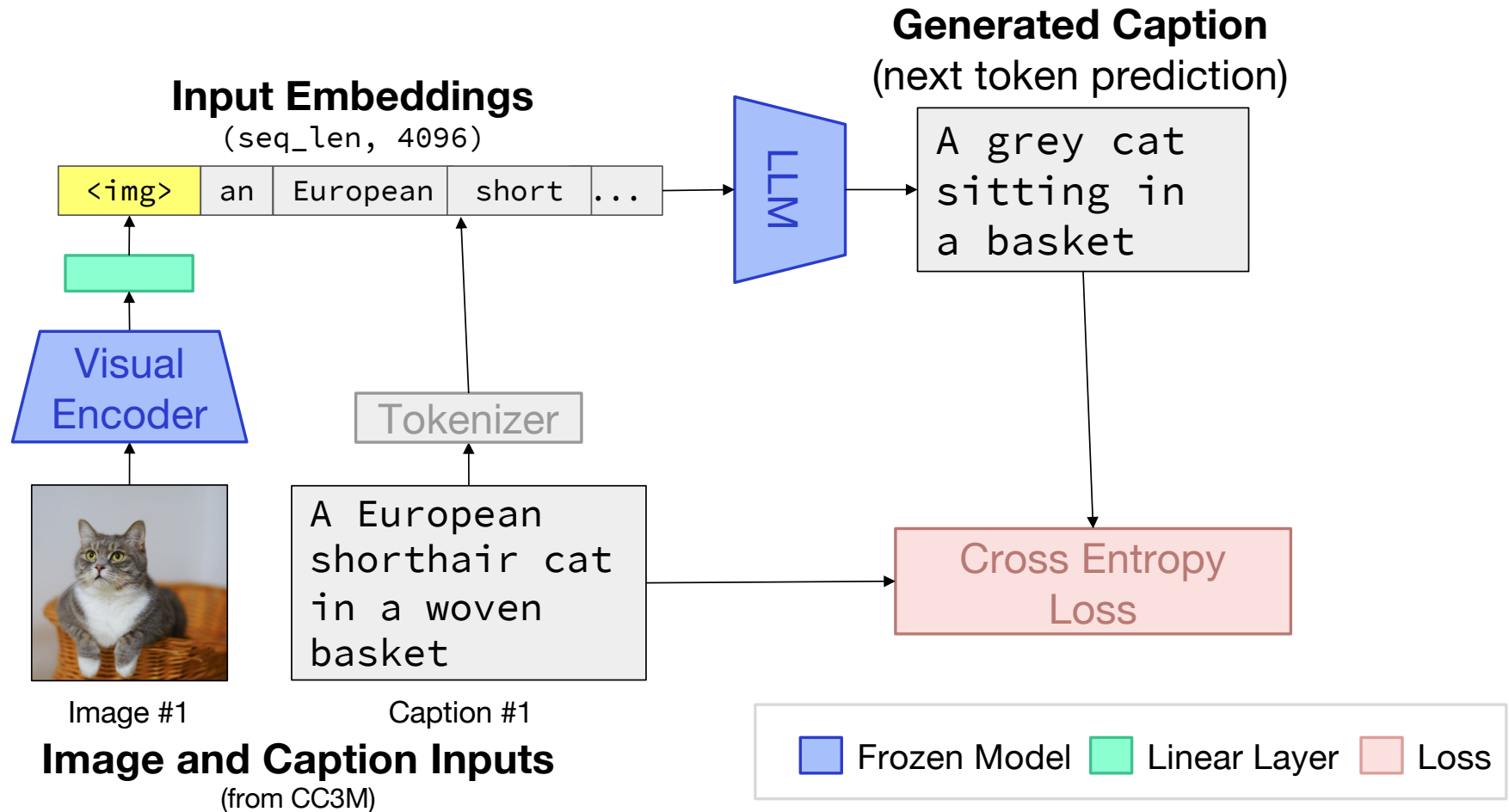
Linearly Mapping from Image to Text Space ([Merullo et al., 2023](#))

Grounding Language Models to Images for Multimodal Inputs and Outputs ([Koh et al., 2023](#))

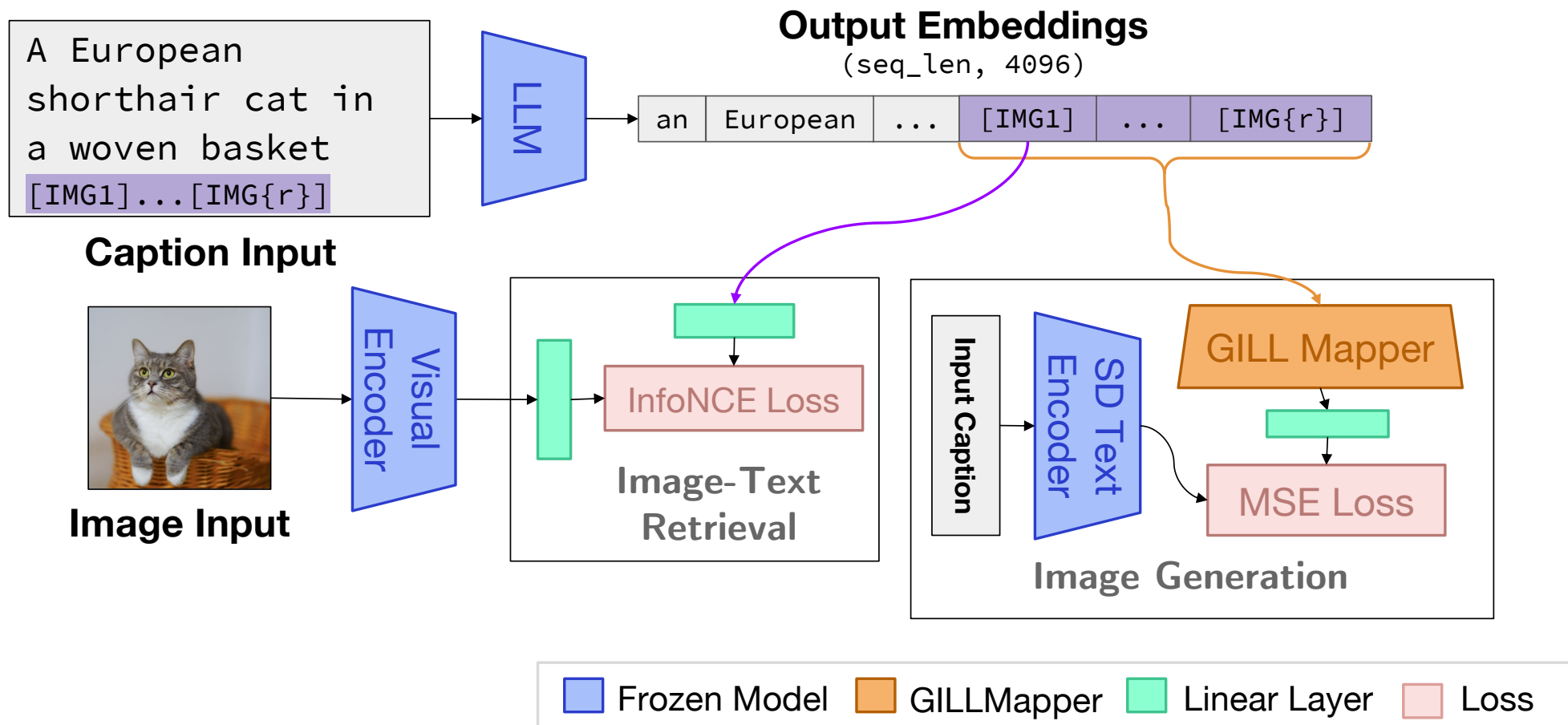
GILL: A More General Multimodal LM

- **Capable of retrieving images, generating images, and generating text**
 - Can condition on arbitrarily interleaved image + text inputs
 - Generate text, generate images, and retrieve images as part of the output
- **Leverage the learnt abilities of pre-trained text-only LLMs**
 - In-context learning
 - Generate long and coherent dialogue
- **Model agnostic**
 - Use a 7B LLM, the CLIP encoder, and the Stable Diffusion image generator
 - Likely benefits from using larger and stronger LLMs in the future

Learning to Process Images

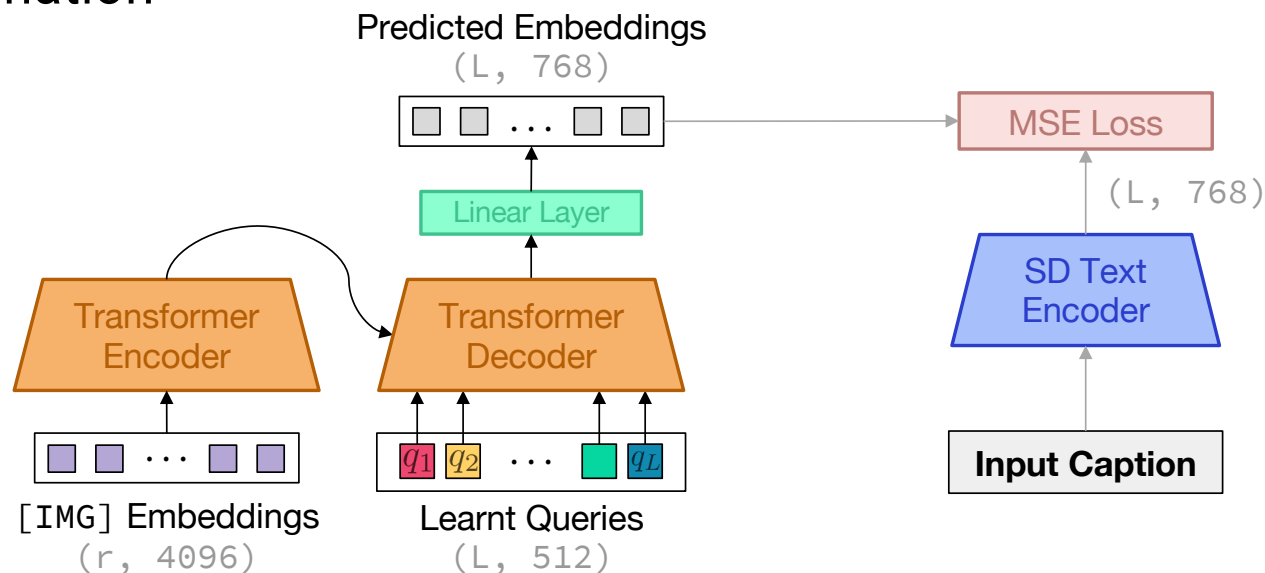


Learning to Produce Images



GILL Mapper: LLM-to-Generator Map

- Many approaches use linear mappings between LLMs and visual models
- This is insufficient for image generation: decoders require dense information

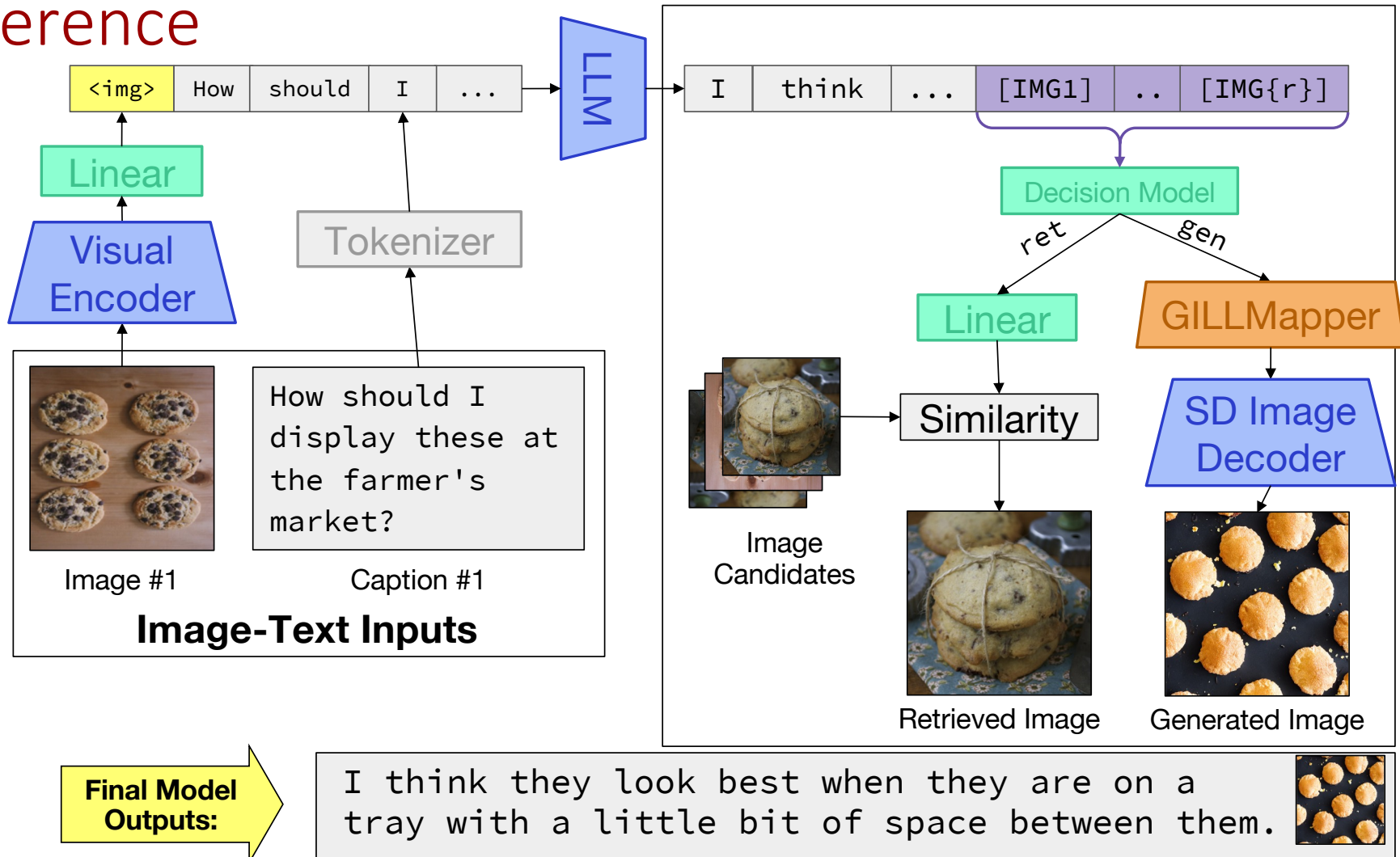


Multimodal Few-Shot Learning with Frozen Language Models (Tsimpoukelli et al., 2021)

Linearly Mapping from Image to Text Space (Merullo et al., 2023)

Grounding Language Models to Images for Multimodal Inputs and Outputs (Koh et al., 2023)

Inference



Evaluation: Contextual Image Generation

- Given a Visual Story, generate a relevant image
- Need to condition on long, temporally dependent text and image inputs

Once while I was on vacation in this nice brick hotel		I woke up and took my dog Trixie for a walk.		Trixie ran around and enjoyed the fresh air.		We had lots of fun playing fetch together		After a while she got tired and had to take a rest.			
Image and Text Inputs											
											
									Stable Diffusion	Ours	Groundtruth

Visual Storytelling ([Huang et al., 2016](#))

Evaluation: Contextual Image Generation

- Given a Visual Story, generate a relevant image
- Need to condition on long, temporally dependent text and image inputs



Model	CLIP Similarity ($\uparrow$)			LPIPS ($\downarrow$)		
	1 caption	5 captions	5 caps, 4 images	1 caption	5 captions	5 caps, 4 images
GLIDE [34]	0.582	0.591	-	0.753	0.745	-
Stable Diffusion [43]	0.592 ± 0.0007	0.598 ± 0.0006	-	0.703 ± 0.0003	0.704 ± 0.0004	-
GILL	0.581 ± 0.0005	0.612 ± 0.0011	0.641 ± 0.0011	0.702 ± 0.0004	0.696 ± 0.0008	0.693 ± 0.0008

- Model outperforms Stable Diffusion on longer input contexts
- GILL benefits from the abilities of the LLM (sensitivity to longer inputs, word orderings, in-context learning)

Visual Storytelling ([Huang et al., 2016](#))

Evaluation: Contextual Image Generation

- Given a Visual Dialogue, generate a relevant image
- Need to condition on long dialogue-like text

Q: is the man alone? A: yes, the man is alone 1	Q: is it sunny outside? A: no, it is not sunny outside 2	Q: what color is the snowboard? A: the snowboard is grey in color 3	Q: is the man wearing a cap? A: the man is wearing a black cap 4	...	Q: what color are the glasses? A: the glasses are white in color 8	Q: can you see the sky? A: no it's totally dark 9	Q: does it look like he's having fun? A: he seems to be enjoying 10	→	  
VisDial Inputs									
Q: what color are the dogs? A: 1 of the dog is white and the other dog is light brown 1	Q: can you tell what breed they are? A: i can't really tell what breed they are, perhaps german shepherd 2	Q: are they both wearing a hat? A: only 1 is wearing a hat 3	...	Q: are they standing in grass? A: no, they are standing on dirt 8	Q: are they looking at each other? A: no, they are facing away from each other 9	Q: do they seem like they like each other? A: can't tell 10	→	  	
VisDial Inputs									Stable Diffusion Ours Groundtruth

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Visual Dialog ([Das et al., 2017](#))

Evaluation: Contextual Image Generation

- Given a Visual Dialogue, generate a relevant image
- Need to condition on long dialogue-like text

Q: is the man alone? A: yes, the man is alone 1	Q: is it sunny outside? A: no, it is not sunny outside 2	Q: what color is the snowboard? A: the snowboard is grey in color 3	Q: is the man wearing a cap? A: the man is wearing a black cap 4	...	Q: what color are the glasses? A: the glasses are white in color 8	Q: can you see the sky? A: no it's totally dark 9	Q: does it look like he's having fun? A: he seems to be enjoying 10			
VisDial Inputs								Stable Diffusion	Ours	Groundtruth

Q: what color are the dogs? A: 1 of the dog is white and the other dog is light brown 1	Q: can you tell what breed they are? A: i can't really tell what breed they are, perhaps german shepherd 2	Q: are they both wearing a hat? A: only 1 is wearing a hat 3	...	Q: are they standing in grass? A: no, they are standing on dirt 8	Q: are they looking at each other? A: no, they are facing away from each other 9	Q: do they seem like they like each other? A: can't tell 10			
VisDial Inputs							Stable Diffusion	Ours	Groundtruth

Model	CLIP Similarity (↑)			LPIPS (↓)		
	1 round	5 rounds	10 rounds	1 round	5 rounds	10 rounds
GLIDE [34]	0.562	0.595	0.587	0.800	0.794	0.799
Stable Diffusion [43]	0.552 ±0.0015	0.629 ±0.0015	0.622 ±0.0012	0.742 ±0.0010	0.722 ±0.0012	0.723 ±0.0008
GILL	0.528 ±0.0014	0.621 ±0.0009	0.645 ±0.0010	0.742 ±0.0022	0.718 ±0.0028	0.714 ±0.0006

The Effect of Context

Multi-modal context is **worth more** than uni-modal context, producing more relevant generation results.

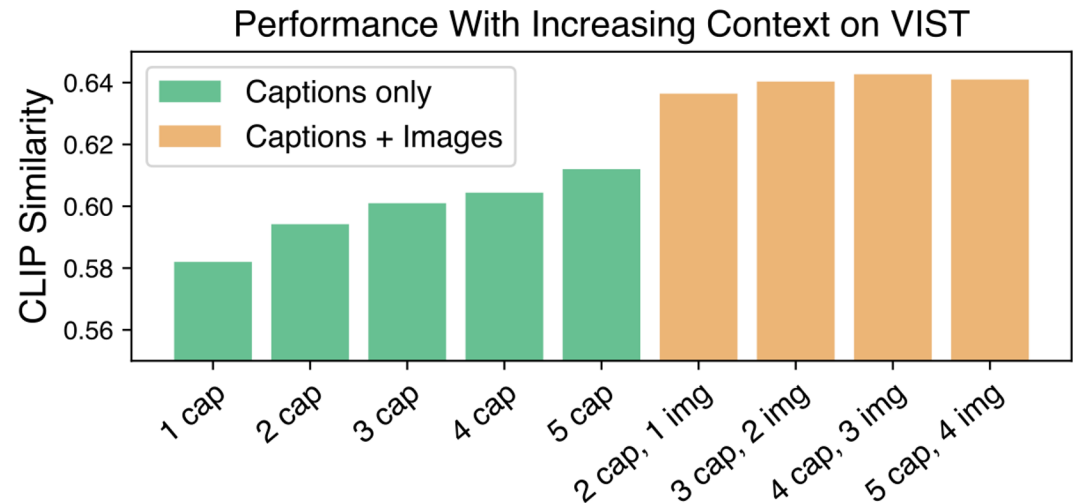


Image Refinement

I want to paint a landscape with mountains and a river. Can you help me find some inspiration?

I'm not sure



Can you show me some with horses?



What about something more abstract?



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Try the Model

huggingface.co/spaces/jykoh/gill

Spaces
 jykoh/gill
 Like 8
 Running on A10G
 Logs

App
 Files
 Community 2
 Settings

GILL

This is the official Gradio demo for the GILL model, a model that can process arbitrarily interleaved image and text inputs, and produce image and text outputs.

Paper: [Generating Images with Multimodal Language Models](#)

Project Website: [GILL Website](#)

Code and Models: [GitHub](#)

Tips:

- Start by inputting either image or text prompts (or both) and chat with GILL to get image-and-text replies.
- Tweak the level of sensitivity to images and text using the parameters on the right.
- Check out cool conversations in the examples or community tab for inspiration and share your own!
- If the model outputs a blank image, it is because Stable Diffusion's safety filter detected inappropriate content. Please try again with a different prompt.
- Outputs may differ slightly from the paper due to slight implementation differences. For reproducing paper results, please use our [official code](#).
- For faster inference without waiting in queue, you may duplicate the space and use your own GPU: [Duplicate Space](#)

GILL Chatbot

How can I publicize these?

I would suggest you start with a local newspaper.



(Generated)

Message

Type a message

Submit

Undo

Reset All

Upload image

Frequency multiplier for returning images (higher means more frequent)

1.3

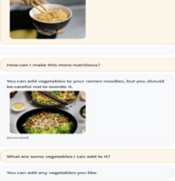
Max # of words

32

Temperature (0 for deterministic, higher for more randomness)

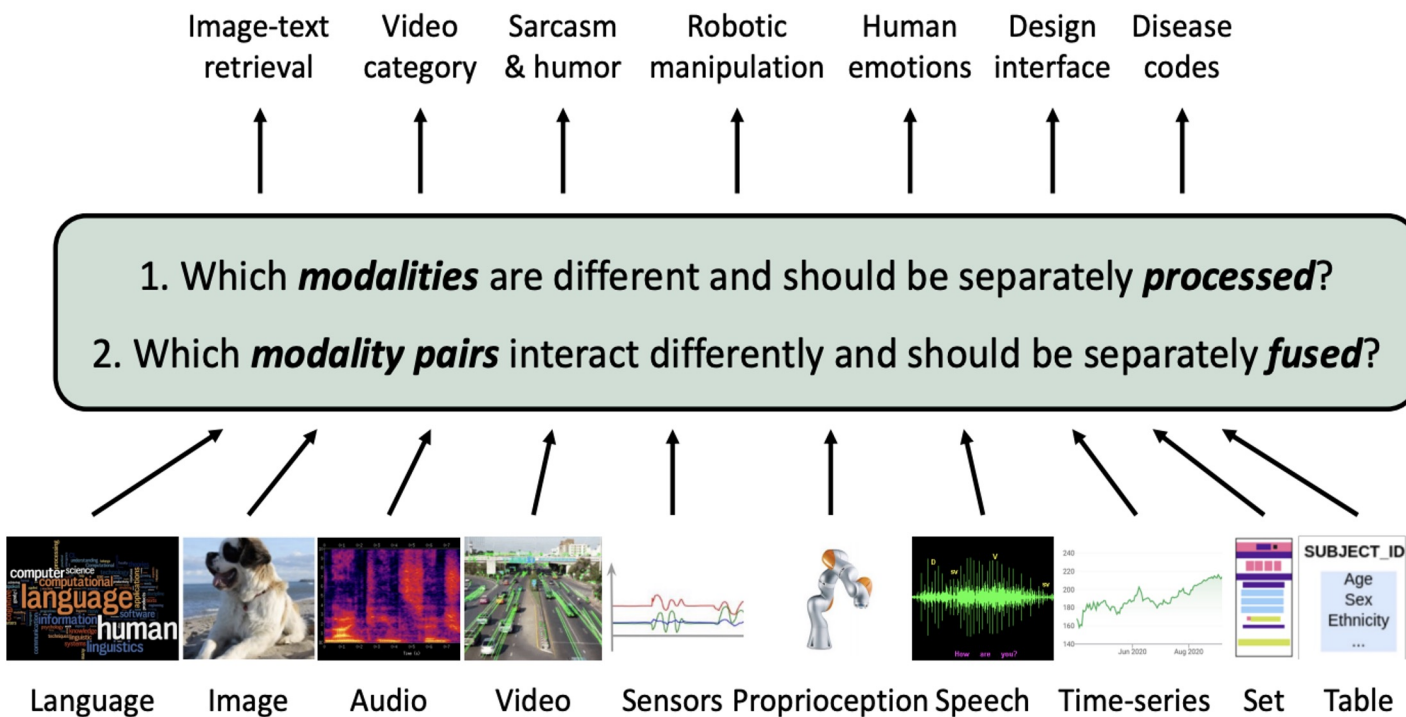
0

Example Conversations



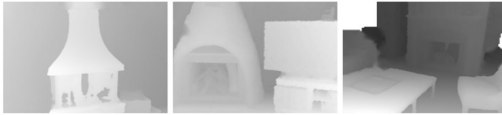



Future Research: Incorporating More Modalities



Future Research: Incorporating More Modalities

1) Cross-Modal Retrieval

Audio	Images & Videos	Depth	Text
 Crackle of a Fire			"A fire crackles while a pan of food is frying on the fire." "Fire is crackling then wind starts blowing." "Firewood crackles then music..."
 Baby Cooing			"A baby is crying while a toddler is laughing." "A baby is laughing while an adult is laughing." "A baby laughs and something..."

2) Embedding-Space Arithmetic


 $\oplus$

 $\rightarrow$


Waves

3) Audio to Image Generation

 Dog
  Engine
  Fire
  Rain

Figure 1. IMAGEBIND’s joint embedding space enables novel multimodal capabilities. By aligning six modalities’ embedding into a common space, IMAGEBIND enables: **1)** Cross-Modal Retrieval, which shows *emergent* alignment of modalities such as audio, depth or text, that aren’t observed together. **2)** Adding embeddings from different modalities naturally composes their semantics. And **3)** Audio-to-Image generation, by using our audio embeddings with a pre-trained DALL-E-2 [61] decoder designed to work with CLIP text embeddings.

Future Research: Long Context Multimodal Models



Gemini

```
[
  {
    "title": "The Lord of the Rings",
    "author": "J.R.R. Tolkien"
  },
  {
    "title": "Structure and Interpretation of Computer Programs",
    "author": "Harold Abelson and Gerald Jay Sussman"
  },
  {
    "title": "Rework",
    "author": "Jason Fried and David Heinemeier Hansson"
  },
  {
    "title": "The Hacker Ethic and the Spirit of the Information Age",
    "author": "Pekka Himanen"
  },
  {
    "title": "The Google Story",
    "author": "David A. Vise"
  },
  ...
]
```

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[The killer app of Gemini Pro 1.5 is video](#)

Future Research: Analysis on Architecture Design

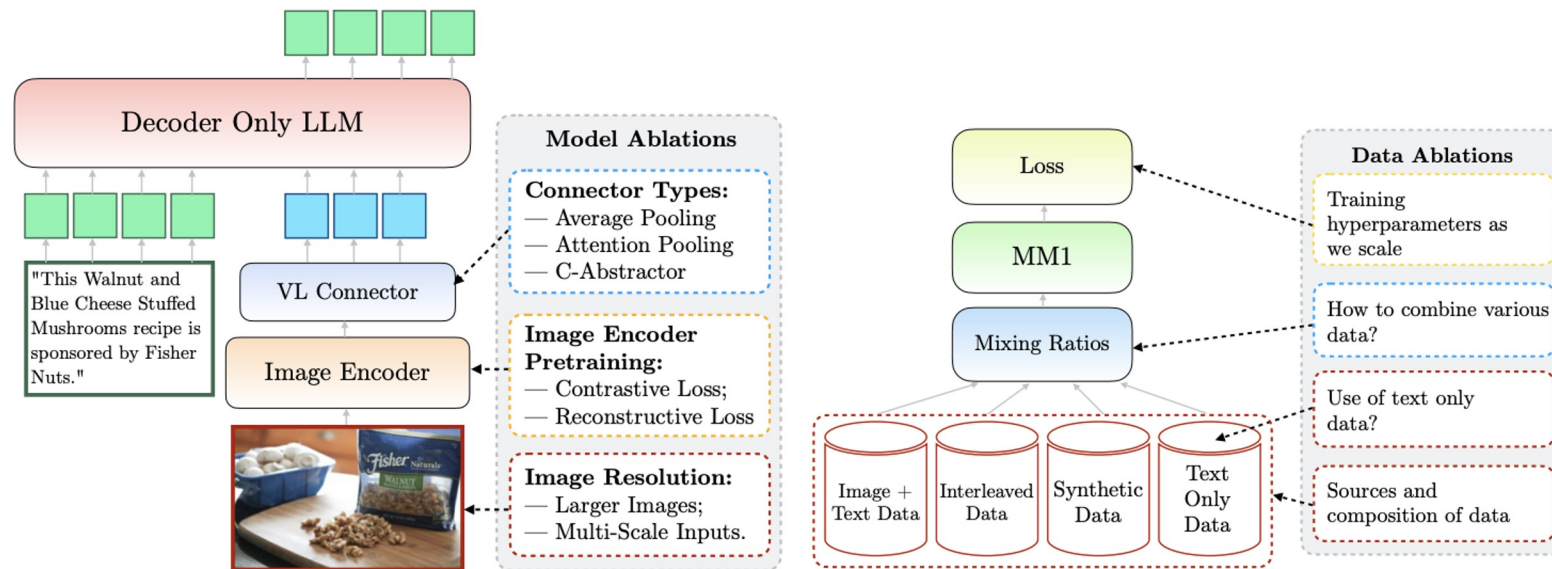
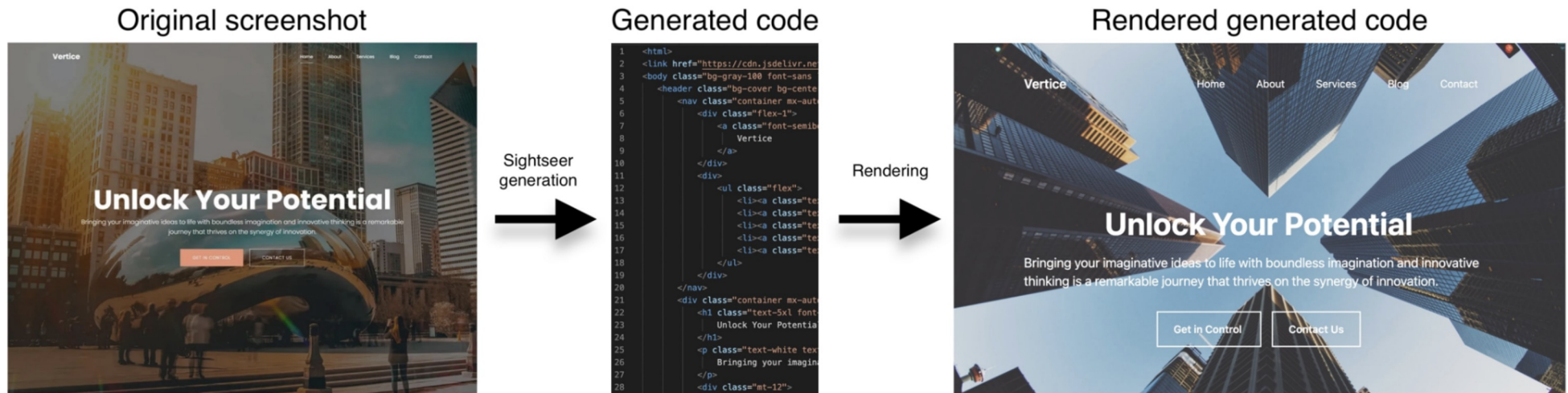


Fig. 3: *Left:* Model ablations: what visual encoder to use, how to feed rich visual data, and how to connect the visual representation to the LLM. *Right:* Data ablations: type of data, and their mixture.

Future Research: Multimodal Code Models



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Unlocking the Conversion of Web Screenshots into HTML Code with the WebSight Dataset ([Laurençon et al., 2024](#))
Design2Code: How Far Are We From Automating Front-End Engineering? ([Si et al., 2024](#))

Other Resources

- **Blog post:** [Multimodality and Large Multimodal Models \(LMMs\)](#)
- **Courses:** [CMU 11-777: Multimodal Machine Learning](#)
 - Lectures available on [YouTube](#)
 - [11-877: Advanced Topics in Multimodal Machine Learning](#)
- **Survey papers:**
 - [Foundations and Trends in Multimodal Machine Learning: Principles, Challenges, and Open Questions](#)
 - [A Survey on Multimodal Large Language Models](#)
 - [Multimodal Large Language Models: A Survey](#)